

Chapter 12 Missing Data

Missing data are common in many research problems. Sometimes missing data arise from design, but more often data are missing for reasons that are beyond researchers' control. I will first provide some conceptual discussion on the types of missing data, and then talk about the Bayesian approach for handling missing data by treating missing data as parameters with some prior information. I will then give a brief introduction of *multiple imputation* and its Bayesian origin. A regression with missing data problem will be used to illustrate two Bayesian approaches to handle missing data.

12.1 Missing Data Mechanisms

To simplify the discussion, assume that missing values are only present in the outcome Y in a hypothetical regression problem of using people's age (X) to predict their voting intention (Y).

Let R be an indicator variable that denotes whether Y is missing ($R = 0$) or not ($R = 1$). For example, if Y looks like

```
># [1] 2.4 2.7 2.4 2.0 2.0 2.0 3.9 3.2 2.3 1.7 NA
```

then R will be

```
># [1] 1 1 1 1 1 1 1 1 1 1 0
```

Assume our data look like the first scatter plot below if there are no missing data:

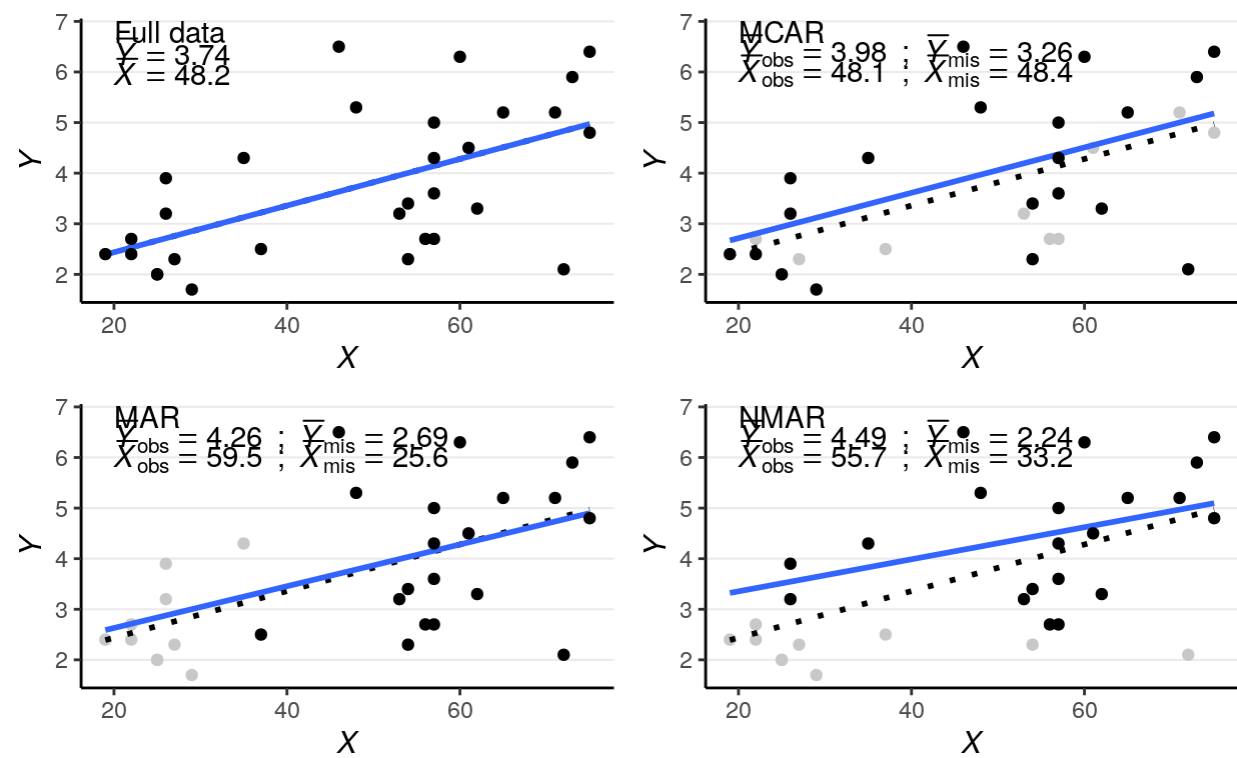
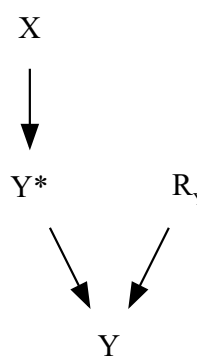


Figure 12.1: Scatter plots for different types of missing data

Missing data can be related to the predictor X (e.g., older people are more likely to give a missing response), the outcome Y itself (e.g., people with lower voting intention are less likely to respond), and some other unmeasured factors that relate to neither X nor Y , which I summarize as Z . Depending on what causes missing data, the three missing data algorithms are $MCAR$ (missing completely at random), MAR (missing at random), and $NMAR$ (not missing at random), as summarized in the figures below, which I will further explain.

12.1.1 MCAR (Missing Completely at Random)



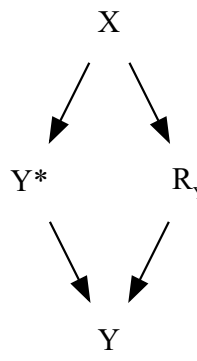
$MCAR$ means that the probability of a missing response (denoted as R) is unrelated to anything of interest in the research question. For example, for the left graph in Figure 2, Z maybe some haphazard events such as interviewers accidentally erase responses for some people, which we believe to be unrelated to participants' ages or voting intentions. The plot on the top right panel of Figure 1 is an example, with the missing cases being grayed out.

One quick-and-dirty method to check for MCAR is to check whether the distribution of X is similar for cases with or without missing data on Y , and as you can see in the above graph the means and variances of X for the group with missing data and for the group without are highly similar. This method can be generalized to data with missing data on multiple variables, and one can check whether missing data on every variable affect all other variables.

As you can see, the regression line barely changes with or without the missing data.

In general, under $MCAR$, using only cases with no missing value still give valid inferences and unbiased estimations. However, for more complex models complete case analyses (also called listwise deletion) can greatly reduce the sample size for analysis, as it throws away information from cases with partial information.

12.1.2 MAR (Missing At Random)

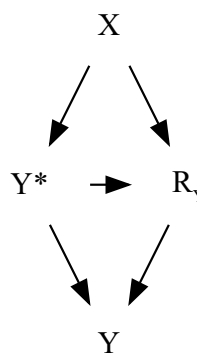


It's probably not the most intuitive naming, but MAR refers to the condition that the probability of a missing observation (R) can be explained by the observed data (i.e., X in this case). In other words, missing data does not relate to the values that would have been observed (which is denoted as Y_{miss}), once we considered the observed data. For example, for the middle graph in Figure 2, some missing data on voting intentions can be explained by some random factor Z , but for some cases data are missing because, for instance, younger people tend to be less motivated to complete the survey. The plot on the bottom left panel of Figure 1 is an example, with the missing cases being grayed out.

As can be seen, when data are MAR, the distributions of X are different for groups with and without missing Y values. Also, the distributions of the observed Y values differ systematically from the complete data.

Under MAR , using only the cases without missing values still produces an unbiased estimate of the regression coefficient, if missing data is only present in Y . However, for more complex models and with missing data in X , more advanced methods generally give more accurate coefficient estimates and standard errors.

12.1.3 NMAR (Not Missing At Random)



$NMAR$ is sometimes called *missing not at random* or *non-ignorable missingness*, and as the name suggested it refers to conditions where MAR does not hold. In other words, NMAR happens when, after considering all the observed data, the probability of a missing value (R) still depends on the value of Y that would have been observed. For example, if we consider people in the same age group and still find those with lower voting intentions tend not to give their responses, the situation can be described as $NMAR$. The plot on the bottom right panel of Figure 1, where people with lowing voting intentions are more likely to miss.

The example looks very similar to the one for MAR, including the fact that the distributions of X are different for the group with and without missing Y . Indeed, there are no statistical procedures that can distinguish between MAR in general and NMAR. If there are evidence for MCAR then one can be more confident in ruling out NMAR, and there have been recent efforts to establish procedures for testing some special cases of MAR. However, for many real data problems one has to rely on reasoning, judgments, and perhaps some educated guessing to decide whether the data is MAR or NMAR.

On the other hand, if one has variables that potentially relates to the probability of missing but are not part of the model of interest (e.g., gender, SES, etc), these can be included in the imputation model (discussed later) so that the missing data mechanism better resembles MAR. Including these *auxiliary* variables is equivalent to changing them from unmeasured to measured, and generally can weaken the associations between the unobserved Y and R , thus making the estimates less biased.

With $NMAR$, valid statistical inferences can only be obtained by correctly modeling the mechanism for the missing data. Including variables that help explain probability of missing data makes $NMAR$ more reasonable.

12.1.4 Ignorable Missingness*

Let Y_{obs} be the part of the multivariate data Y that is observed (i.e., not missing), and Y_{miss} be the part that would have been observed. The likelihood now concerns both Y_{obs} and R , that is, $P(Y_{obs}, R)$. Let ϕ be the set of parameters that determine the probability of missing in addition to the observed data, which can be written as $P(R|Y_{obs}, \phi)$. Note it is assumed that ϕ is distinct from the model parameters θ .

For a case i with $r_i = 1$, the joint likelihood of $(x_i, y_i, r_i = 1)$ is

$$P(x_i, y_{obs,i}, r_i = 1; \theta, \phi) = P(r_i = 1 | x_i, y_{obs,i}; \phi) P(y_{obs,i} | x_i; \theta) P(x_i).$$

For a case with $r_i = 0$, y_i is missing. Assume first we know the missing value $y_{miss,i}$, and the complete likelihood $(x_i, y_{miss,i}, r_i = 0)$ is

$$P(x_i, y_{miss,i}, r_i = 0; \theta, \phi) = P(r_i = 0 | x_i, y_{miss,i}; \phi) P(y_{miss,i} | x_i; \theta) P(x_i)$$

But because y_i is missing, we need to integrate out the missing value to obtain the observed likelihood of $(x_i, r_i = 0)$

$$\begin{aligned} P(x_i, r_i = 0; \theta, \phi) &= \int P(r_i = 0 | x_i, y_{miss,i}; \phi) P(y_{miss,i} | x_i; \theta) P(x_i) dy_{miss,i} \\ &= P(x_i) \int P(r_i = 0 | x_i, y_{miss,i}; \phi) P(y_{miss,i} | x_i; \theta) dy_{miss,i} \end{aligned}$$

Because the likelihood depends on R and cannot be separated from ϕ , correct inference on θ can be obtained only by correct modeling the missing data mechanism.

12.1.4.1 If MCAR Holds

However, if the condition for MCAR is satisfied such that

$$P(r_i = 0|x_i, y_{\text{mis},i}; \phi) = P(r_i = 0; \phi),$$

that is, R is related to neither X and Y Then the observed likelihood is

$$\begin{aligned} P(x_i, r_i = 0; \theta, \phi) &= P(x_i) \int P(r_i = 0; \phi) P(y_{\text{mis},i}|x_i; \theta) dy_{\text{mis},i} \\ &= P(x_i) P(r_i = 0; \phi) \times \int P(y_{\text{mis},i}|x_i; \theta) dy_{\text{mis},i} \\ &= P(x_i) P(r_i = 0; \phi) \end{aligned}$$

So inference of θ does not depend on the missing data mechanism $P(r_i = 0; \phi)$, and missingness is *ignorable*.

12.1.4.2 If MAR Holds

Similarly, if the condition for MAR is satisfied such that

$$P(r_i = 0|x_i, y_{\text{mis},i}; \phi) = P(r_i = 0|x_i; \phi),$$

that is, R is not related to Y after taking into account X . Then the observed likelihood is

$$\begin{aligned} P(x_i, r_i = 0; \theta, \phi) &= P(x_i) \int P(r_i = 0|x_i; \phi) P(y_{\text{mis},i}|x_i; \theta) dy_{\text{mis},i} \\ &= P(x_i) P(r_i = 0|x_i; \phi) \times \int P(y_{\text{mis},i}|x_i; \theta) dy_{\text{mis},i} \\ &= P(x_i) P(r_i = 0|x_i; \phi) \end{aligned}$$

So inference of θ does not depend on the missing data mechanism $P(r_i = 0|x_i; \phi)$, and missingness is ignorable.

On the other hand, if r_i depends on $y_{\text{mis},i}$ (i.e., NMAR) so that $P(r_i = 0|x_i, y_{\text{mis},i}; \phi)$ cannot be written outside of the integral, inference of θ depends on the missing data mechanism, so missingness is non-ignorable.

The discussion generalizes to missing data on multiple variables.

12.2 Bayesian Approaches for Missing Data

We will be using the `kidiq` data set we discussed in Chapter 7. I'll do the same rescaling and coding `mom_hs` as a factor variable:

```
kidiq <- haven::read_dta("../data/kidiq.dta")
kidiq100 <- kidiq %>%
  mutate(mom_iq = mom_iq / 100, # divid mom_iq by 100
         kid_score = kid_score / 100, # divide kid_score by 100
         mom_iq_c = mom_iq - 1,
         mom_hs = factor(mom_hs, labels = c("no", "yes"))) %>%
  select(- mom_iq)
```

In R, the package `mice` can be used to perform *multiple imputation* (to be discussed soon), as well as to create missing data. First, let's generate some missing completely at random (MCAR) data by randomly removing up to 50% of the data:

```
library(mice)
set.seed(1955)
kidiq100_mcar <- ampute(kidiq100, prop = 0.5,
                        pattern = data.frame(kid_score = c(0, 0, 0),
                                             mom_hs = c(1, 1, 0),
                                             mom_work = c(1, 1, 1),
                                             mom_age = c(1, 1, 1),
                                             mom_iq_c = c(1, 0, 1)),
                        freq = c(.2, .4, .4),
                        mech = "MCAR")
kidiq100_mcar <- kidiq100_mcar$samp
```

The second time, I'll generate some missing at random (MAR) data:

```
set.seed(1955)
kidiq100_mar <- ampute(kidiq100, prop = 0.5,
                      pattern = data.frame(kid_score = c(0, 0, 0),
                                           mom_hs = c(1, 1, 0),
                                           mom_work = c(1, 1, 1),
                                           mom_age = c(1, 1, 1),
                                           mom_iq_c = c(1, 0, 1)),
                      freq = c(.2, .4, .4),
                      mech = "MAR")
```

```
># Warning: Data is made numeric because the calculation of weights requires
># numeric data
```

```
kidiq100_mar <- kidiq100_mar$samp
```

And finally, some not missing at random (NMAR) data:

```
set.seed(1955)
kidiq100_nmar <- ampute(kidiq100, prop = 0.5,
                       pattern = data.frame(kid_score = c(0, 0, 0),
                                             mom_hs = c(1, 1, 0),
                                             mom_work = c(1, 1, 1),
                                             mom_age = c(1, 1, 1),
                                             mom_iq_c = c(1, 0, 1)),
                       freq = c(.2, .4, .4),
                       # mice call it MNAR
                       mech = "MNAR")
```

```
># Warning: Data is made numeric because the calculation of weights requires
># numeric data
```

```
kidiq100_nmar <- kidiq100_nmar$samp
```

Let's check the distributions of the resulting data:

```
># 'geom_smooth()' using method = 'loess' and formula 'y ~ x'
```

```
># Warning: Removed 230 rows containing non-finite values (stat_smooth).
```

```
># Warning: Removed 230 rows containing missing values (geom_point).
```

```
># 'geom_smooth()' using method = 'loess' and formula 'y ~ x'
```

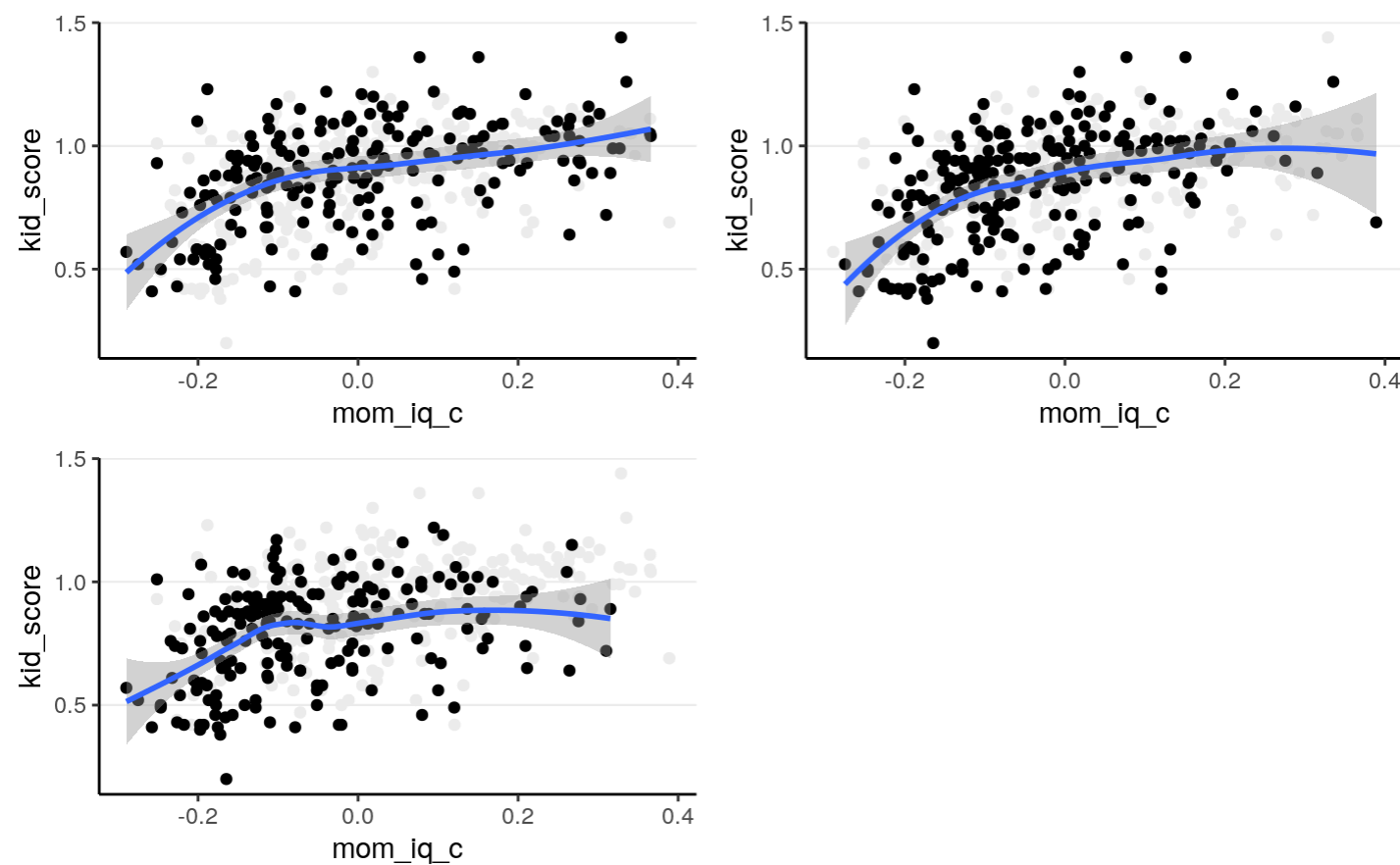
```
># Warning: Removed 215 rows containing non-finite values (stat_smooth).
```

```
># Warning: Removed 215 rows containing missing values (geom_point).
```

```
># 'geom_smooth()' using method = 'loess' and formula 'y ~ x'
```

```
># Warning: Removed 239 rows containing non-finite values (stat_smooth).
```

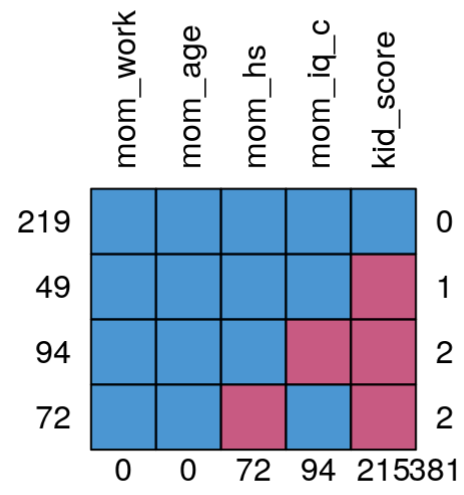
```
># Warning: Removed 239 rows containing missing values (geom_point).
```



When eyeballing it doesn't appear that the data are very different, but the regression slopes are affected by the different missing data mechanisms. We'll look at the simple regression model of using `mom_iq_c` to predict `kid_score`, using the MAR data set. In that data set, the missingness of `kid_score` actually depends on both `mom_iq_c` and `mom_hs`, but when the regression does not include `mom_hs` in the model, the resulting situation will actually be NMAR.

The missing data pattern of the `kidiq100_mar` data set is:

```
# Recode mom_hs to factor
kidiq100_mar$mom_hs <- factor(kidiq100_mar$mom_hs, labels = c("no", "yes"))
md.pattern(kidiq100_mar, rotate.names = TRUE)
```

```
>#      mom_work mom_age mom_hs mom_iq_c kid_score
># 219      1      1      1      1      1      0
># 49      1      1      1      1      1      0      1
># 94      1      1      1      0      0      0      2
># 72      1      1      0      1      1      0      2
>#      0      0      72      94      215 381
```

Which shows that only 219 observations had full data, and most were missing the `kid_score` variable.

12.2.1 Complete Case Analysis/Listwise Deletion

By default, `brms` uses only cases with no missing data. For example,

```
m3_id <- brm(kid_score ~ mom_iq_c + mom_hs, data = kidiq100_mar,
  prior = c(prior(normal(0, 1), class = "Intercept"),
    # set for all "b" coefficients
    prior(normal(0, 1), class = "b"),
    prior(student_t(4, 0, 1), class = "sigma")),
  seed = 2302
)
```

```
># Warning: Rows containing NAs were excluded from the model.
```

```
># Family: gaussian
># Links: mu = identity; sigma = identity
># Formula: kid_score ~ mom_iq_c + mom_hs
># Data: kidiq100_mar (Number of observations: 219)
># Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
>#           total post-warmup samples = 4800
>#
># Population-Level Effects:
>#           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
># Intercept      0.81      0.03    0.76    0.86 1.00   3395   3055
># mom_iq_c        0.71      0.11    0.50    0.92 1.00   3518   2641
># mom_hsys        0.07      0.03    0.01    0.13 1.00   3583   3274
>#
># Family Specific Parameters:
>#           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
># sigma      0.20      0.01    0.18    0.22 1.00   3632   2916
>#
># Samples were drawn using sampling(MUTS). For each parameter, Eff.Sample
># is a crude measure of effective sample size, and Rhat is the potential
># scale reduction factor on split chains (at convergence, Rhat = 1).
```

Notice that the number of observations is only 219. As previously explained, this analysis is only valid when data are missing completely at random or missing at random (i.e., missingness of the outcome only depends on `mom_iq_c` and factors unrelated to `ozone`).

If you recall in Chapter 7, the coefficient using the full data should be:

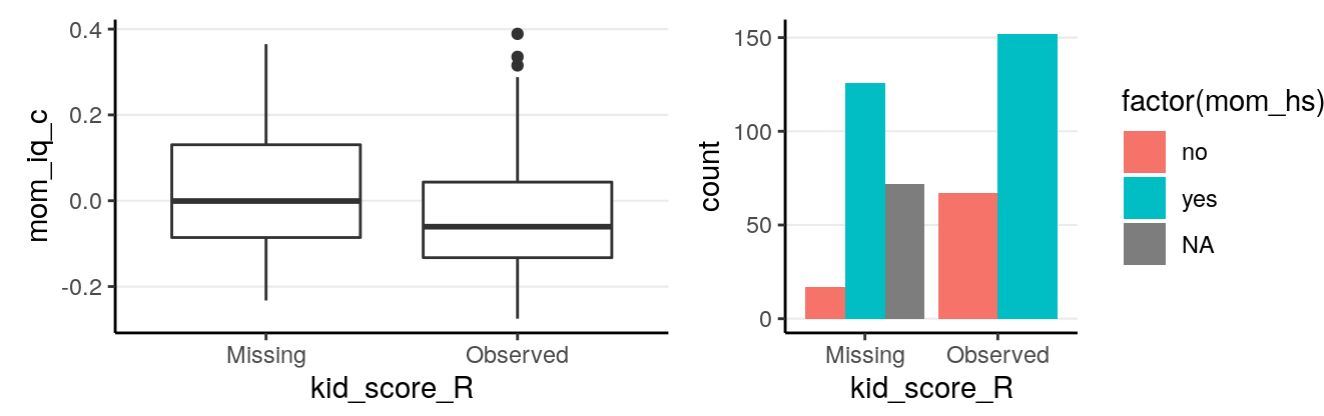
```
># Family: gaussian
># Links: mu = identity; sigma = identity
># Formula: kid_score ~ mom_iq_c + mom_hs
># Data: kidiq100 (Number of observations: 434)
># Samples: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
>#           total post-warmup samples = 4800
>#
># Population-Level Effects:
>#           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
># Intercept      0.82      0.02    0.78    0.86 1.00   3840   3093
># mom_iq_c        0.56      0.06    0.44    0.68 1.00   3672   2883
># mom_hsys        0.06      0.02    0.02    0.10 1.00   3761   3213
>#
># Family Specific Parameters:
>#           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
># sigma      0.18      0.01    0.17    0.19 1.00   3899   3136
>#
># Samples were drawn using sampling(MUTS). For each parameter, Eff.Sample
># is a crude measure of effective sample size, and Rhat is the potential
># scale reduction factor on split chains (at convergence, Rhat = 1).
```

So the listwise approach overestimated the regression coefficient. We can do better.

Now, take a look on whether missingness in `kid_score` is related to other variables.

```
# Compute the missingness indicator (you can use the 'within' function too)
kidiq100_mar_R <- transform(kidiq100_mar,
  kid_score_R = factor(as.numeric(!is.na(kid_score)),
    labels = c("Missing",
      "Observed")))
```

```
># Warning: Removed 94 rows containing non-finite values (stat_boxplot).
```



As we already knew, missingness of `kid_score` is related to both `mom_iq_c` and `mom_hs`, in that those with higher `mom_iq_c` and those whose mother had high school degree were more likely to be missing.

12.2.2 Treat Missing Data as Parameters

A fully Bayesian approach to handle missing data is to treat the missing `kid_score` values just as parameters, and assign priors to them. When the missing data mechanism is ignorable (MCAR or MAR), we can assume that the missing and observed `kid_score` values are exchangeable, conditioning on the predictors (i.e., whether `kid_score` is missing or not does not add information to the `kid_score` values). Therefore, if `kid_score` is missing, we use the likelihood as the prior for the missing values:

$$\begin{aligned} \text{kid_score}_{\text{obs},i} &\sim \mathcal{N}(\beta_0 + \beta_1 \text{mom_iq}_{ci}, \sigma) \\ \text{kid_score}_{\text{mis},i} &\sim \mathcal{N}(\beta_0 + \beta_1 \text{mom_iq}_{ci}, \sigma) \\ \beta_0 &\sim \mathcal{N}(0, 1) \\ \beta_1 &\sim \mathcal{N}(0, 1) \\ \beta_2 &\sim \mathcal{N}(0, 1) \end{aligned}$$

```
data {
  int<lower=0> N; // number of observations
  vector[N] y; // response variable (including missing values);
  int<lower=0, upper=1> y_obs[N]; // missingness indicator for Y
  int<lower=0> p; // number of predictor variables (exclude intercept)
  matrix[N, p] X; // predictor variable matrix
}

transformed data {
  int n_obs = sum(y_obs); // number of observed cases
  int ns[n_obs]; // indices of observed cases
  int ny = 1;
  for (n in 1:N) {
    if (y_obs[n]) {
      ns[ny] = n;
      ny += 1;
    }
  }
}

parameters {
  real beta_0; // intercept
  vector[2] beta; // 2 slopes
  real<lower=0> sigma; // error standard deviation
}

model {
  // likelihood for observed Y
  y[ns] ~ normal_id_glm(X[ns, ], beta_0, beta, sigma);
  // prior
  beta_0 ~ normal(0, 1);
  beta ~ normal(0, 1);
  sigma ~ student_t(4, 0, 1);
}

generated quantities {
  real yrep[N]; // simulated data based on model
  vector[N] yhat = beta_0 + X * beta; // used to compute R-squared effect size
  for (i in 1:N) {
    yrep[i] = normal_rng(yhat[i], sigma);
  }
}
```



```
m3_stan <- stan("../codes/normal_regression_missing.stan",
  data = list(N = nrow(kidiq100_mar_obsX),
    y = replace_na(kidiq100_mar_obsX$kid_score, 99),
    y_obs = as.numeric(
      is.na(kidiq100_mar_obsX$kid_score)
    ),
    p = 2,
    X = model.matrix(~ mom_iq_c + mom_hs,
      data = kidiq100_mar_obsX[, ~1]),
  seed = 1234)
```

Note that the results are basically identical to the complete case analyses, and the posterior distributions of the missing Y values are essentially the predictive intervals given the X values. For example, for the first 10 observations with missing `kid_score` values,

```
draws_ymls <-
  as.matrix(m3_stan, pars = "yrep")[ , is.na(kidiq100_mar_obsX$kid_score)]
mcmc_areas_ridges(draws_ymls[, 1:10], bw = "SJ")
```

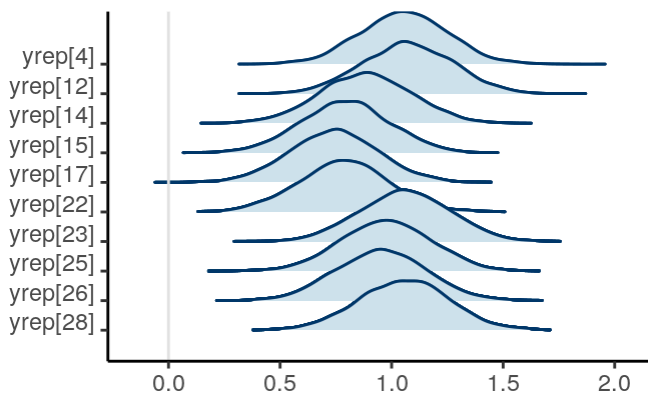


Figure 12.2: Posterior density plots of the first two missing values of `texttt(kid_score)`

The posterior distributions of the missing values are highly related to the missing data handling technique called *multiple imputation*, which we will discuss next. Indeed, each posterior sample can be considered an imputed data set. The posterior draws of the missing values are also called *plausible values*.

12.2.3 Multiple Imputation

Multiple imputation is one of the modern techniques for missing data handling, and is general in that it has a very broad application. It uses the observed data and the observed associations to predict the missing values, and captures the uncertainty involved in the predictions by imputing multiple data sets. That's a bit abstract, but with your Bayesian knowledge, that just means *getting samples from the posterior distributions of the missing values*, and then substitute them to the missing holes to form an imputed data set. The difference is that, instead of using all posterior samples, we usually obtain 20 or 30 imputed data sets, which can be saved and used for almost any kind of analyses, Bayesian or frequentist.

12.2.3.1 Multiple imputation has several advantages

- It provides valid results when data is MAR
- It reduces biases when data is NMAR by incorporating covariates that help explain the missing data mechanism (e.g., `mom_work` and `mom_age`)
- It is very flexible and can impute continuous and categorical variables

12.2.3.2 Example of multiple imputation

Although in theory one can use the Bayesian procedures with `Stan` to account for missing data or to do multiple imputations, there are some limitations. First, when the goal is to impute missing data instead of making inferences on the model parameters, the algorithm in `Stan` may not be as efficient as specialized programs for multiple imputation. Second, the Hamiltonian Monte Carlo sampler in `Stan` requires the use of derivatives, so it is not (yet) well-equipped to handle categorical parameters. Thus, it is hard or not possible to handle categorical missing data. Third, when the number of variables with missing data is large, it is tedious to specify the missing data mechanism for all variables.

Instead, as Gelman et al. (2013) recommended, we can handle missing data using a *two-step process*:

1. Do multiple imputation using a specialized program
2. Use `brms` or `rstan` (or other Bayesian methods) to analyze each imputed data set

12.2.3.3 R packages for multiple imputation

There are several packages in R for multiple imputation (e.g., `Amelia`, `jomo`, `mi`, `mice`, `missForest`, `norm`, `pan`). Although these packages differ in terms of their underlying algorithms, my experience and also evidence from the literature suggested that they usually gave similar performance for continuous missing data, but several packages have specialized functionality for specific models and data types (e.g., categorical missing data, multilevel data).

I will illustrate the use of `mice` below. I strongly encourage you to take a look on the vignettes found on the website of the package: <https://github.com/stefvanbuuren/mice>. Also, the author of the package has a nice book on multiple imputation (Van Buuren 2018), which is freely available at <https://stefvanbuuren.name/fimd/> and I encourage you to read if you are interested. Note that the example discussed here is simple so not much fine tuning for the imputation is needed. For your own analyses multiple imputation can be complex, and you should consult statisticians or other resources to set up a reasonable imputation model.

Let's continue with the `kidiq` example. We can use the whole data set for imputation.

In general it's recommended to include covariates that have even minor associations with the probability of missing. The bias introduced by ignoring an important covariate usually is higher than the bias introduced by including an inappropriate covariate. However, see Thoemmes and Rose (2014) for a cautionary note.

In planning a study, if high missing rate on a variable is anticipated, one can collect covariates that can help explain the missing data mechanism. This helps recover missing information in the analyses.

12.2.3.3.1 1. Setting up and run the imputation

With binary and continuous missing variables, it can be as simple as running the following:

```
library(mice)
# Using mice to impute 20 data sets
kidiq100_imp <- mice(kidiq100_mar, m = 20, maxit = 35,
  printFlag = FALSE) # set to false only for knitting to Rmd
```

Of course this oversimplifies the complexity of multiple imputation. By default it uses the method called "predictive mean matching" to replace missing data with a randomly chosen value from several similar cases (see <https://stefvanbuuren.name/fimd/sec-pmm.html>). Things will get more complicated when you have more variables and complex data types.

Typing `kidiq100_imp$imp` will show the imputed missing values. Check `?mice` for more information. The `complete` function fills the missing values to the missing holes to form data sets with no missing data.

```
kidiq100_imp1 <- complete(kidiq100_imp, 1)
head(kidiq100_imp1, 10)
```

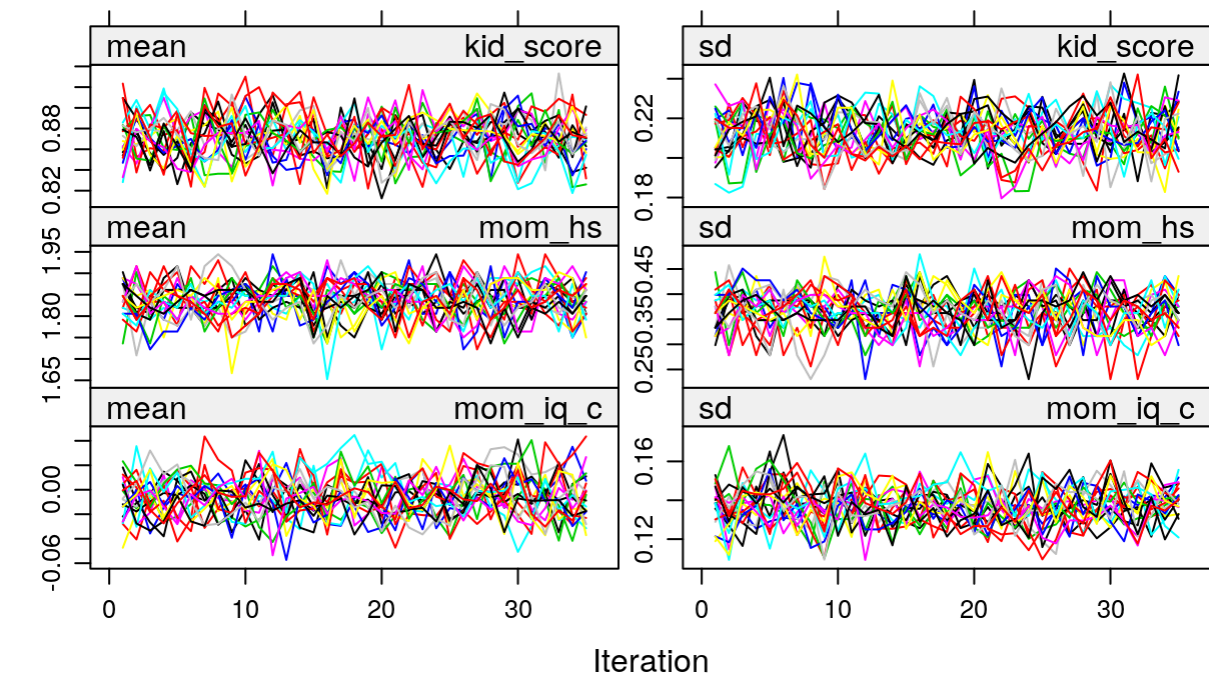
```
>#   kid_score mom_hs mom_work mom_age mom_iq_c
># 1    1.21   yes      4      27   -0.1883
># 2    1.26   yes      4      25    0.2754
># 3    0.85   yes      4      27    0.1544
># 4    0.41   yes      3      25   -0.1785
># 5    0.42   yes      4      27   -0.0725
># 6    0.98   no       1      18    0.0790
># 7    0.69   yes      4      20    0.3889
># 8    1.16   yes      3      23    0.2515
># 9    1.02   yes      1      24   -0.1838
># 10   0.95   yes      1      19   -0.0493
```

Compared to the original data:

```
># # A tibble: 10 x 5
>#   kid_score mom_hs mom_work mom_age mom_iq_c
>#   <dbl> <fct>   <dbl>   <dbl>   <dbl>
># 1    0.65 yes      4      27    0.211
># 2    0.98 yes      4      25   -0.106
># 3    0.85 yes      4      27    0.154
># 4    0.83 yes      3      25   -0.00550
># 5    1.15 yes      4      27   -0.0725
># 6    0.98 no       1      18    0.0790
># 7    0.69 yes      4      20    0.389
># 8    1.06 yes      3      23    0.251
># 9    1.02 yes      1      24   -0.184
># 10   0.95 yes      1      19   -0.0493
```

12.2.3.3.2 2. Check for Convergence

We should also look at convergence:



These are basically Markov chains in regular Bayesian analyses. So if you see some chains are constantly above or below others then it's problematic.

See https://www.gerkovink.com/mice/Vignettes/Convergence_pooling/Convergence_and_pooling.html for additional steps to check for convergence.

12.2.3.3.3 3. Run brm_multiple on imputed data sets

brms directly supports multiply imputed data sets. Simply use the `brm_multiple` function and supply the multiply imputed data object to it. Also, for computational efficiency using two chains for each imputed data set would be faster.

```
m3_imp <- brm_multiple(kid_score ~ mom_iq_c + mom_hs, data = kidiq100_imp,
  prior = c(prior(normal(0, 1), class = "Intercept"),
    # set for all "b" coefficients
    prior(normal(0, 1), class = "b"),
    prior(student_t(4, 0, 1), class = "sigma")),
  seed = 1955,
  chains = 2L,
  cores = 2L
)
```

See this vignette: https://cran.r-project.org/web/packages/brms/vignettes/brms_missings.html#compatibility-with-other-multiple-imputation-packages for more information. If you look at the results:

```
># Warning: The model has not converged (some Rhats are > 1.05). Do not analyse the results!
># We recommend running more iterations and/or setting stronger priors.
```

```
># Family: gaussian
># Links: mu = identity; sigma = identity
># Formula: kid_score ~ mom_iq_c + mom_hs
># Data: kidiq100_imp (Number of observations: 434)
># Samples: 40 chains, each with iter = 2000; warmup = 1000; thin = 1;
>#          total post-warmup samples = 40000
>#
># Population-Level Effects:
>#           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
># Intercept      0.80      0.02    0.76    0.85 1.15    165    813
># mom_iq_c        0.67      0.09    0.48    0.85 1.39     85    332
># mom_hsyes       0.07      0.03    0.02    0.13 1.18    145    599
>#
># Family Specific Parameters:
>#           Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
># sigma      0.19      0.01    0.18    0.21 1.20    133    237
>#
># Samples were drawn using sampling(NUTS). For each parameter, Eff.Sample
># is a crude measure of effective sample size, and Rhat is the potential
># scale reduction factor on split chains (at convergence, Rhat = 1).
```

You will see that there are 40 chains in the results. The `Rhat` value will be much higher than 1, as the chains are from different data sets and will never converge. Instead, you should investigate the `Rhat` for each data set by

```
>#      b_Intercept b_mom_iq_c b_mom_hsyes sigma lp__
># 1      1.000      1.003      1.000 0.999 1.000
># 2      1.001      1.000      1.002 1.001 1.000
># 3      1.000      1.000      1.000 1.000 1.001
># 4      0.999      0.999      0.999 1.000 1.001
># 5      0.999      1.000      0.999 0.999 1.001
># 6      0.999      1.000      0.999 1.001 0.999
># 7      1.000      1.000      1.000 0.999 1.000
># 8      0.999      0.999      1.000 0.999 1.000
># 9      0.999      1.000      1.000 0.999 1.001
># 10     1.000      1.001      1.000 1.002 0.999
># 11     1.000      0.999      1.000 1.001 1.002
># 12     0.999      1.003      0.999 1.001 1.000
># 13     1.001      1.000      1.000 1.001 1.001
># 14     0.999      0.999      0.999 0.999 1.001
># 15     1.000      1.000      1.001 0.999 1.001
># 16     1.000      0.999      1.000 0.999 1.004
># 17     1.001      0.999      1.002 0.999 1.000
># 18     1.000      0.999      1.000 1.000 1.000
># 19     0.999      0.999      1.000 1.000 1.001
># 20     1.000      1.001      1.000 1.000 1.006
```

So the chains have converged for each individual data set.

Now, put the results together:

```
># Warning: The model has not converged (some Rhats are > 1.05). Do not analyse the results!
># We recommend running more iterations and/or setting stronger priors.
```

```
># Warning: Using only the first imputed data set. Please interpret the results
># with caution until a more principled approach has been implemented.
```

```
>#
># =====
>#           Full data      Complete case MI
># -----
># Intercept      0.82 *      0.81 *      0.80 *
>#           [0.78; 0.86] [0.76; 0.86] [0.76; 0.85]
># mom_iq_c       0.56 *      0.71 *      0.67 *
>#           [0.43; 0.67] [0.48; 0.90] [0.48; 0.85]
># mom_hsyes      0.06 *      0.07 *      0.07 *
>#           [0.02; 0.10] [0.01; 0.13] [0.02; 0.13]
># -----
># R^2           0.21      0.22      0.22
># Num. obs.    434      219      434
># =====
># * 0 outside the confidence interval
```

You can see that the coefficients for `mom_iq_c` is closer to the original data with multiple imputation, and the credible intervals are slightly shorter than complete case analyses.

For data with more variables, choices of missing data handling method can make a substantial difference. Therefore, researchers need to be thoughtful in choosing imputation models that best reflect the missing data mechanism.

Missing data is an active research area, and this note only covers a very small fraction of the issues discussed in the literature.

12.3 References

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1. Price is another important figure in mathematics and philosopher, and have taken Bayes' theorem and applied it to insurance and moral philosophy.↗
 2. See the [paper by John Aldrich](#) on this.↗
 3. See <http://plato.stanford.edu/entries/probability-interpret/> for more information↗
 4. In a purely subjectivist view of probability, assigning a probability P to an event does not require any justifications, as long as it follows the axioms of probability. For example, I can say that the probability of me winning the lottery and thus becoming the richest person on earth tomorrow is 95%, which by definition would make the probability of me not winning the lottery 5%. Most Bayesian scholars, however, do not endorse this version of subjectivist probability, and require justifications of one's beliefs (that has some correspondence to the world).↗
 5. The likelihood function in classical/frequentist statistics is usually written as $P(y; \theta)$. You will notice that here I write the likelihood for classical/frequentist statistics to be different than the one used in Bayesian statistics. This is intentional: In frequentist conceptualization, θ is fixed and it does not make sense to talk about probability of θ . This implies that we cannot condition on θ , because conditional probability is defined only when $P(\theta)$ is defined.↗

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